

FAIRMONT, MN, USA

WMO: 726586

Lat: 43.650N

Lon: 94.417W

Elev: 1161

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94948

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-6.8	-22.1	1.7	-9.4	-15.4	2.5	-5.3	34.4	13.8	31.8	10.9	10.5	330	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	90.0	74.3	87.5	73.0	83.8	70.9	77.4	85.6	75.3	83.1	73.4	81.2	15.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.8	135.9	83.1	72.6	126.0	81.0	70.5	117.2	78.7	41.6	85.4	39.5	83.3	37.7	81.8	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
30.2	26.7	24.2	-17.3	94.7	5.7	3.0	-21.4	96.9	-24.7	98.7	-27.8	100.4	-32.0	102.6	
			-17.5	80.7	5.4	2.2	-21.4	82.3	-24.5	83.5	-27.5	84.8	-31.5	86.3	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	16.0	20.0	32.6	46.3	58.8	69.0	72.0	69.2	62.6	49.1	34.5	21.4
	DBStd	22.09	12.97	12.74	12.31	9.87	8.71	6.41	5.48	5.45	8.21	9.28	11.10	11.88
	HDD50	4154	1055	841	553	181	21	0	0	0	6	132	477	888
	HDD65	7551	1520	1261	1006	565	231	32	8	20	141	499	915	1353
	CDD50	2724	0	0	12	69	295	570	683	596	384	103	12	0
	CDD65	648	0	0	0	3	41	152	226	152	68	6	0	0
	CDH74	5648	0	0	4	56	464	1337	1947	1126	647	65	2	0
	CDH80	1682	0	0	0	12	150	432	610	285	181	12	0	0
	WSAvg	11.4	12.9	12.9	12.7	13.4	12.9	10.6	8.4	7.7	9.5	11.3	12.6	12.5
	PrecAvg	32.7	0.8	0.8	1.9	3.3	4.2	4.6	4.3	4.2	3.4	2.3	1.7	1.0
(15) Precipitation	PrecMax	52.4	2.3	3.1	4.9	6.3	9.1	14.5	11.5	15.2	10.2	8.0	5.7	4.3
	PrecMin	19.6	0.0	0.0	0.2	0.3	1.3	1.5	0.8	0.7	0.6	0.1	0.1	0.1
	PrecStd	6.3	0.6	0.6	1.1	1.6	2.0	2.4	2.3	2.8	2.3	1.9	1.4	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	54.2	70.4	81.3	90.4	92.9	91.5	90.4	90.0	81.2	69.7	54.1
		MCWB	39.5	47.7	55.6	60.9	68.8	72.3	77.7	76.4	71.3	65.3	54.9	48.5
	2%	DB	38.9	45.4	62.7	73.1	83.8	88.5	89.9	86.2	84.4	73.5	62.6	45.6
		MCWB	35.1	40.9	53.5	56.9	64.1	71.6	76.8	74.8	69.7	60.2	52.4	40.2
	5%	DB	36.2	39.7	55.1	67.9	79.3	84.3	86.3	82.4	81.4	69.7	55.2	39.2
		MCWB	33.4	36.3	48.5	54.0	61.6	69.0	74.6	72.3	66.9	58.1	47.7	35.5
	10%	DB	33.8	36.5	48.4	63.0	73.5	81.5	82.5	80.8	77.4	63.9	50.1	36.3
		MCWB	31.6	33.7	43.7	51.5	59.0	67.0	72.1	70.5	66.0	54.3	44.3	33.6
	0.4%	WB	39.7	48.7	60.1	63.2	71.8	77.1	81.4	79.5	74.9	70.5	57.9	49.6
		MCDB	45.6	53.5	65.5	75.6	84.6	87.9	89.0	87.5	84.4	76.4	66.2	54.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.6	41.2	54.5	59.3	67.8	74.4	78.1	76.4	72.2	63.1	53.0	40.4
		MCDB	38.5	45.4	60.8	69.5	78.8	84.2	86.2	83.7	81.3	70.1	60.5	45.8
	5%	WB	33.4	37.0	49.5	55.5	64.6	71.9	76.1	74.3	69.9	59.3	48.4	36.3
		MCDB	35.7	40.2	55.2	65.4	74.7	81.8	83.8	81.1	78.1	66.9	54.8	39.6
	10%	WB	31.3	33.8	44.2	52.5	61.8	69.4	74.2	71.9	67.4	55.2	44.4	33.4
		MCDB	33.1	36.4	49.1	60.9	70.8	78.4	81.8	78.4	74.6	63.5	49.7	35.9
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	15.8	15.9	19.3	19.7	18.9	18.7	18.5	21.6	19.6	16.5	14.3
		MCDBR	17.2	18.0	25.0	29.9	27.1	23.4	21.1	20.5	25.4	27.2	24.9	19.4
		MCWBR	14.3	14.8	17.2	16.9	13.0	11.1	11.7	12.1	12.8	15.6	16.5	15.4
	5% WB	MCDBR	15.9	17.1	23.0	24.9	22.3	19.9	18.8	18.5	21.5	22.7	23.1	18.4
		MCWBR	13.6	14.1	16.6	16.0	12.5	11.7	11.8	12.0	12.4	14.9	16.1	14.9
(40) Clear-Sky Solar Irradiance	taub		0.271	0.285	0.313	0.365	0.384	0.406	0.417	0.409	0.367	0.327	0.286	0.264
	taud		2.437	2.408	2.440	2.328	2.345	2.332	2.281	2.290	2.376	2.486	2.523	2.479
	Ebn at Noon		277	291	295	285	282	275	270	268	272	272	269	270
	Edh at Noon		23	28	31	38	39	39	41	39	33	26	21	20
(44) All-Sky Solar Radiation	RadAvg		580	914	1204	1480	1737	1922	2056	1708	1372	907	601	442
	RadStd		63	105	122	143	124	149	87	123	91	119	50	40

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page